Test membranes, step by step

Panama Blot Analyzer 0.2.1 | November 2026

Start with **02-warped-groups** in **experiments/Test-pack**. It contains 18 simulated samples, their Western (W) and protein-stain (P) TIFFs, and a known eight-band pattern. These are numerical test fixtures, not biological evidence.

Open a finished result

Double-click **Open test experiment.cmd**, or use **Load experiment** and choose **02-warped-groups/02-analyzed.panama**. Open **06 Analysis**. Select PCA, Heatmap, LDA, K-means or Statistics. The saved experiment includes all five results. Use **Save as...** for your own copy.

Reproduce from the TIFF images

1. Import. Click New. In Images & lanes, import the file ending **-W.tif** as the blot and **-P.tif** as protein stain. Alternatively load **01-images-only.panama**, which contains only those images. Keep Dark intensity mode.

2. Place lanes. Set **Lane count = 18**, then Detect lanes. Inspect W and P overlays. Protein strips lie between sample lanes; the final two samples share the last strip. Mark lanes reviewed if prompted. Keep the full height; do not auto-crop this benchmark.

3. Align. Open Alignment. Set Protein lane to **Sample 01**. Click Auto reference, then Match lanes. There should be **8 landmarks**, with no unmatched flags. Review positions, then click Apply alignment.

4. Set background. In Baseline, choose **Constant**, Level **0.06**, check Apply to all lanes, then Apply baseline. This is the known synthetic background.

5. Set sections. In Sections, choose **Template... > Load template** and select this case's **sections.json**. It restores the exact eight integration intervals used for validation.

6. Label samples. In Analysis > Samples..., set samples 01-06 to **Group 1**, 07-12 to **Group 2**, and 13-18 to **Group 3**. Leave Pair blank. Apply.

7. Run. Use Area, Normalize None, Scale Mean center, log off. Run PCA and Heatmap. Run K-means with **K = 3, seed = 42**, LDA with seed 42, and Statistics with Welch ANOVA.

8. Save. Click **Excel...** to choose where to save the matrix and results. Save experiment stores the editable work. Closing offers **Save & close**, **Discard changes** or **Cancel** when needed.

Auto setup remains available for exploration. To reproduce these exact expected numbers, use the explicit baseline, full-height crop and section template above.

What should the tests show?

01-calibration | 12 samples

Repeat steps 1-5 with Lane count **12**. No groups or pairs are needed. Every sample has the same pattern multiplied by a known gain. PC1 should explain essentially **100%** of variance; its sign is arbitrary. Maximum area error is **0.0026%**. This checks signal extraction and quantitative integration.

02-warped-groups | 18 samples

PCA should explain **49.86% on PC1** and **48.93% on PC2**. K-means should recover the three planted groups exactly (cluster numbers may differ). LDA cross-validation accuracy is **100%** on this deliberately easy simulated example. It does not establish real biological prediction performance.

Median pairwise protein-profile correlation rises from **0.596 to 0.999** after alignment. Maximum band-area error is **0.390%**; detected lane centers differ from the known positions by at most **1 pixel**.

03-paired-change | six matched pairs

Use Lane count **12** and the same alignment/background/template steps. In Samples..., label 01-06 **Before** and 07-12 **After**. Assign pairs P01-P06 to each block: samples 01 and 07 share P01, 02 and 08 share P02, and so on.

Choose Statistics > Paired t test, **Group A = After**, **Group B = Before**. Band 03 has a planted amplitude increase of 0.10. Its BH-adjusted q is approximately **3.43 x 10^-8**. Maximum area error is **0.380%**. Pair matching follows IDs, not row position. LDA is not run for these repeated subjects.

How to check the numbers

expected-areas.csv contains independent analytic Gaussian integrals. Compare them with the band columns in the Excel **Measurements** sheet. These comparisons require Area, None normalization and log off. Error (%) = 100 x |measured - expected| / expected.

ground-truth.json records the random seed, amplitudes, positions, spatial profile, background and imposed migration shifts. **validation.json** records each measured check. The builder imports actual 16-bit TIFF files through the same image and analysis commands used by the interface.

Each case contains an images-only experiment and an analyzed experiment. **results.xlsx** is a saved reference workbook. Loading an analyzed experiment recomputes its saved analyses. These controlled cases validate implemented calculations under stated conditions; they do not cover every scanner, staining artifact or biological design.

Try a published immunoblot

Vale, Nobrega and collaborators (2013)

Open **published/04-Vale-2013-serum/02-exploratory.panama**. This uses the 15 serum lanes from supplementary Figure S1A: five DeltaD-iD, five WT, and five DeltaD-DFL. The TIFF is an unchanged-pixel rectangular crop of the figure's embedded JPEG.

Inspect: Images & lanes shows narrow ROIs inside the displayed strips. Alignment is skipped because the supplement does not provide a matching Protogold image. Baseline uses Rolling 10%, window 121. Auto sections proposes 17 intervals. PCA, heatmap and descriptive statistics are saved as exploratory outputs.

Repeat: load 01-image-only.panama or import Vale-2013-S1A-W.tif into a new experiment. Draw 15 W lanes using the crop X positions and width 13 in sample-layout.csv. Set each lane from Y = 0 to 486. Set the same baseline and load sections.json. Apply the listed genotype labels in Samples..., then run PCA, Heatmap and descriptive Statistics.

With these practice choices, PC1 is **37.30%** and PC2 is **27.17%**. These are Panama results for the published display pixels, not published PCA results.

What can and cannot be reproduced

The S1 caption states that the serum lanes were spliced from different locations in the same immunoblot. The image is compressed, prepared for publication and lacks raw intensity calibration, the protein scan and the authors' numeric profiles. It is useful for extraction, manual editing and exploratory analysis, but cannot verify exact reproduction of their measurements or conclusions.

The article uses **LDA** to mean **limiting dilution analysis**. Panama's LDA means **linear discriminant analysis**; these are different procedures. Do not interpret the synthetic classification test as a replication of the paper's LDA.

Sources and attribution

Vale AM et al. (including Alberto Nobrega). *The link between antibodies to OxLDL and natural protection against pneumococci depends on DH gene conservation.* J Exp Med 210:875-890 (2013). doi:10.1084/jem.20121861. Figure S1 and caption are retained in the published folder.

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Haury M, Grandien A, Sundblad A, Coutinho A, Nobrega A. *Global analysis of antibody repertoires. 1. An immunoblot method for the quantitative screening of a large number of reactivities.* Scand J Immunol 39:79-87 (1994). doi:10.1111/j.1365-3083.1994.tb03343.x. Its double-stain migration correction motivates the paired W/P synthetic test; no raw scans from this paper were obtained.